

create_pg_special_pattern

NAME

create_pg_special_pattern
Creates a power ground special pattern. This pattern can be instantiated by set_pg_strategy command to create power ground shapes inside the specified routing area.

SYNTAX

```
status create_pg_special_pattern
pattern_name
[-insert_channel_straps spec]
[-insert_terminal_alignment_straps spec]
[-terminal_alignment_via_rule via_rule_spec]
[-insert_power_switch_alignment_straps spec]
[-insert_physical_cell_alignment_straps spec]
[-honor_max_stdcell_strap_distance spec]
[-parameters var_list]
```

Data Types

pattern_name	collection
spec	specification
via_rule_spec	specification
var_list	specification

ARGUMENTS

pattern_name
Specifies the name of the special pattern.

-insert_channel_straps spec
Specifies the channel strap insertion option specification. Channel straps are inserted inside channels formed by objects if no existing straps are inside the channels. The spec specification is as follows:

```
{
{layer : layer_name}
{direction : horizontal | vertical}
{width : width_value}
{spacing : spacing_value}
{offset : offset}
{reference : offset_reference}
{channel_threshold : threshold}
{track_alignment : track | half_track | none}
{channel_between_objects :
    macro | placement_blockage | voltage_area |
    macro_with_keepout | region_boundary | block}
{check_one_layer : true | false}
}
```

The layer keyword followed by the layer_name layer specifies the layer for channel strap insertion; this setting is required. The direction keyword followed by horizontal or vertical specifies the channel direction; the default is vertical. The width keyword followed by width_value specifies the channel strap width; the default is the minimum width. The spacing keyword followed by spacing_value specifies the spacing between channel straps; the default is the minimum spacing. The offset keyword followed by offset specifies the offset from reference to the center of the first strap. The reference keyword followed by offset_reference specifies the offset reference. The offset_reference could be either left or center. Left means channel left boundary. Center means the centerline of the channel. The default is left. However, if the offset is 0, the reference will be changed to center. The channel_threshold keyword followed by threshold specifies the lower width bound for narrow channels. Any channels with width smaller than the threshold are not considered for insertion. The default threshold is 0. The track_alignment keyword followed by track, half_track or none specifies how straps align to the track.

The channel_between_objects keyword followed by the object name specifies the objects considered for channels such as macros, hard placement blockages, voltage areas, macros with hard keepout, routing area boundary, and blocks. When macro_with_keepout is specified, if a hard keepout is not defined for a certain macro, the macro boundary is used for channel detection. When

voltage_area is specified, channels between embedded voltage areas and disjoint voltage areas are supported, while channels between partially overlapped voltage areas are not supported. If no object is specified using the channel_between_objects keyword, by default the tool considers only macros for channels.

The check_one_layer keyword followed by true or false specifies when the tool inserts straps in the channel. If check_one_layer is false, channel straps are not inserted if any straps at any layers are inside the channel. If check_one_layer is true, channel straps are not inserted only if any straps at the specified layer are inside the channel. By default, check_one_layer is false.

`-insert_terminal_alignment_straps spec`

Specifies how terminal alignment straps are inserted. If spec is "on", terminal alignment straps will be inserted for all terminals at all layers. By default, the terminal alignment straps are created at all layers. Terminal straps are trimmed and snapped to shapes in different directions when cut by blockages or DRCs. If dangling wires are preferred, you can turn off the trim option. If specific layers are preferred, layers can be specified in the spec. The spec for trim is described as follows:

```
{trim : true | false}
{layers : layerList}
```

When trim is false, terminal alignment straps are not trimmed. layers is the keyword for layers and layerList specifies the list of layers for terminal alignment strap insertion.

`-terminal_alignment_via_rule via_rule_spec`

Specifies the via rule via_rule_spec between terminal alignment straps. The via rule defines the intersection locations and the via inserted at the locations. This option can be applied only for terminal alignment pattern.

There are several ways to define the intersection locations in the via_rule_spec specification, where the syntax is defined as follows:

```
{intersection : all} {via_def} |
{intersection : adjacent} {via_def} |
list_of_filter_rules |
```

The first method is "{intersection : all}", where intersection is the keyword and all refers to all intersections between any two orthogonal wires in any different layers.

The second method is "{intersection : adjacent}", where intersection is the keyword and adjacent refers to all intersections between any two orthogonal wires only in adjacent layers.

The third method is to define the intersections based on two sets of wire shapes using list_of_filter_rules, which contains a list of filtering rules. Each filtering rule is specified inside a curly brace pair. The syntax of each filtering rule is as follows:

```
{filter_1}{filter_2}{via_def} |
{intersection : undefined}{via_def}
```

where filter_1 refers to the first filtering operation and filter_2 refer to the second filtering operation. The intersection is defined between the first set and second set of wires based on the filtering rules. The second filtering rule is optional, and the intersection is defined based on the first set of wires to all other terminal alignment straps.

The filtering operation is defined as follows:

```
{layers : layer_list} {width : {lower_w upper_w}}
```

where the layers keyword followed by layer_list specifies the wire layer names. The width keyword and {lower_w upper_w} setting specifies the lower and upper range of possible wire widths. For those intersections which do not belong to any filtering rule, the via definition can be specified as follows:

```
{intersection : undefined}{via_def}
```

where intersection is the keyword and undefined refers to the remaining intersections which do not belong to any filtering rules. By default, no via is created at the intersections which do not belong to any filtering rules unless specified. Multiple via filtering rules, such as intersection location and via definition, can be defined and separated by curly braces where each filter rule is in one brace pair.

With the intersection definition, the via specification {via_def} is in the following format:

```
{via_master : via_master_list | via_rule_list}
```

where via_master is the keyword for via masters including via contact code defined in the technology file or user-defined via cells. The PG via rules defined by set_pg_via_master_rule command can be specified in the list. via_master_list specifies the list of via masters. Via masters are created at the intersection center without replication. If offset or specific array dimension is needed, via rules specified by set_pg_via_master_rule command can be used. via_rule_list is a list of PG via rules. NIL can be used as a keyword indicate that no via would be created. default can be used to describe the default vias in the specified location. Multiple via rules can be specified and separated by brace pairs. If NIL is not specified in the list, default contact code will be used to complete a stacked via or to replace a via with DRCs when applicable.

By default, the default via defined in the technology file will be created at each intersection of orthogonal terminal alignment straps in different layers.

-insert_power_switch_alignment_straps spec
Specifies the power switch alignment strap insertion option. The alignment straps are inserted to align with power switch primary power input pins. The spec is defined as follows:

```
{  
{lib_cells : lib_cell_name_list}  
{layer : layer_name}  
{width : width_value}  
{direction : horizontal | vertical}  
{offset : offset_value}  
{pitch : pitch_value}  
{track_alignment : track | half_track | none }  
{number : number}  
{pin_layers : layer_name_list}  
{via_master : via_master_list}  
}
```

The lib_cells keyword followed by lib_cell_name_list specifies the lib cell name list for power switch cells; this option is required. The layer keyword followed by layer_name specifies the layer for strap insertion, this option is required. The width keyword followed by width_value specifies the strap width; this option is optional. By default, the pin width is used. The direction option followed by horizontal or vertical specifies the alignment strap direction, the default is vertical. The offset keyword followed by offset_value specifies the offset between strap center and pin shape center. By default, the offset is 0. The pitch keyword followed by pitch_value specifies the pitch between alignment strap center. The track_alignment keyword followed by track, half_track or none specifies how straps align track. The number keyword followed by number specifies the number of alignment straps for each pin shape. By default, the number is 1. The pin_layers keyword followed by layer_name_list specifies a list of layer names. Only pin shapes on the specified layers are aligned. By default, pins on all layers are aligned. The via_master keyword followed by via_master_list specifies the via masters used for via creation between the alignment straps and pins. via_master_list can include contact codes or design vias defined in the technology file, or via master rule defined by set_pg_via_master_rule command. It can also include default or NIL keywords. By default, the default contact codes are used for via creation.

-insert_physical_cell_alignment_straps spec
Specifies the physical cell alignment strap insertion option. Supported cell types include physical_only, well_tap, and filler. The alignment straps are inserted to align with physical cell pins. The spec is defined as follows:

```
{
{lib_cells : lib_cell_name_list}
{layer : layer_name}
{width : width_value}
{direction : horizontal | vertical}
{offset : offset_value}
{track_alignment : track | half_track | none }
{number : number}
{pin_names : pin_name_list}
{pin_layers : layer_name_list}
{via_master : via_master_list}
}
```

The `lib_cells` keyword followed by `lib_cell_name_list` specifies the `lib cell name` list for physical cells. The `layer` keyword followed by `layer_name` specifies the layer to use for strap insertion; this setting is required. The `width` keyword followed by `width_value` specifies the strap width. This setting is optional. By default, the tool uses the pin width. The `direction` option followed by `horizontal` or `vertical` specifies the alignment strap direction. By default, the direction is vertical. The `offset` keyword followed by `offset_value` specifies the offset between strap center and pin shape center, by default, the offset is 0. The `track_alignment` keyword followed by `track`, `half_track` or `none` specifies how straps align track. The `number` keyword followed by `number` specifies the number of alignment straps for each pin shape. By default, the number is 1. The `pin_names` keyword followed by `pin_name_list` specifies a list of pin names for alignment. By default all pins are considered for alignment. The `pin_layers` keyword followed by `layer_name_list` specifies a list of layer names and only pin shapes on the specified layers are aligned. By default, pins on all layers are aligned. The `via_master` keyword followed by `via_master_list` specifies the via masters used for via creation between the alignment straps and pins. `via_master_list` can include contact codes or design vias defined in the technology file, or via master rule defined by `set_pg_via_master_rule` command. It can also include default or NIL keywords. By default, the default contact codes are used for via creation.

`-honor_max_stdcell_strap_distance spec`

Specifies the extra strap insertion to honor maximum standard cell rail tail distance option. Extra vertical straps are inserted for rail tails longer than the specified maximum distance. The spec is defined as follows:

```
{
{layer : layer_name}
{width : width_value}
{max_distance : threshold}
{offset : offset_list}
{check_layers : layer_name}
}
```

The `layer` keyword followed by `layer_name` specifies the layer for strap insertion; this setting is required. The `max_distance` keyword followed by `threshold` specifies the maximum length of standard cell rail tails; this setting is required. The `width` keyword followed by `width_value` specifies the strap width. This setting is optional; by default, the tool uses the minimum width for the layer. The `check_layers` keyword followed by `layer_name` specifies target layers for standard cell rail tails distance calculation. User can specify one or multiple layers. This is an optional setting. By default, it will check all layers.

The `offset` keyword followed by `offset_list` specifies a list of offset values. This setting is optional. By default, extra straps are inserted such that the distance between the inserted strap center and the nearest strap center is threshold. If `offset_list` is specified, the largest value from `offset_list` is used to calculate the extra strap distance such that the extra strap is within the rail tail range.

`-parameters var_list`

Specifies the list of parameters for this pattern. Each parameter can be evaluated and used as the argument for other options in this command. This allows the pattern to be programmable and reused by passing different parameters in different situations. Each parameter `var` in the list `var_list` is a string. Parameter strings in `var_list` for the `-parameters` option do not contain the `@` character, but the `@` character is used as a prefix symbol for parameters when used with the other command options.

DESCRIPTION

Creates a power ground special pattern. This pattern can be instantiated by the set_pg_strategy command to create special power ground shapes inside the specified routing area.

The special pattern can specify channel strap insertion, terminal alignment strap insertion, or extra strap insertion to honor maximum standard cell rail tail distance. Only one type can be specified for one special pattern.

EXAMPLES

The following example creates a special pattern for channel strap insertion. Vertical channels straps are inserted on layer M8 with width 1.0um. Macros, voltage areas and placement blockages are considered for channels.

```
prompt> create_pg_special_pattern channelPattern \
-insert_channel_straps { \
  {layer: M7}{direction: vertical}{width: 1} \
  {channel_between_objects: {macro placement_blockage voltage_area}} \
}
```

The following example creates a special pattern for terminal alignment strap insertion. The vias between adjacent layers are specified in the terminal alignment via rule.

```
prompt> create_pg_special_pattern terminalPattern \
-insert_terminal_alignment_straps on \
-terminal_alignment_via_rule { \
  {{layers: M8}{layers: M7}{via_master: VIA78}} \
  {{layers: M7}{layers: M6}{via_master: VIA67}} \
}
```

The following example creates a special pattern for power switch cell pin alignment strap insertion. The alignment straps are inserted on layer M4 for power switches with lib cell name HEAD16DM.

```
prompt> create_pg_special_pattern pwrSwitchAlignmentPattern \
-insert_power_switch_alignment_straps { \
  {lib_cells: HEAD16DM} {layer: M4} \
}
```

The following example creates a special pattern for physical cell pin alignment strap insertion. The alignment straps are inserted on layer M4 for tap cells with lib cell names TAP_REF1 and TAP_REF2.

```
prompt> create_pg_special_pattern \
phyCellAlignmentPattern \
-insert_physical_cell_alignment_straps { \
  {lib_cells: TAP_REF1 TAP_REF2} {layer: M4} \
}
```

The following example creates a special pattern for extra strap insertion to honor maximum standard cell rail tail distance. The tool inserts extra straps of width 2um on layer M8, the rail tail maximum length is 20um, the location is decided by offset value list {20 10}.

```
prompt> create_pg_special_pattern \
extraStrapPattern \
-honor_max_stdcell_strap_distance { \
  {layer: M8}{max_distance: 20}{width: 2}{offset: {20 10}} \
}
```

In addition to the above examples, see the Examples page in the PG Planning section of the Task Assistant for more information. To view the Examples page, start the GUI and invoke the following command: gui_show_task_assistant -task "Design Planning:PG Planning->Examples->Overview".

SEE ALSO

[compile_pg\(2\)](#)
[remove_pg_patterns\(2\)](#)
[report_pg_patterns\(2\)](#)
[report_pg_strategies\(2\)](#)
[set_pg_strategy\(2\)](#)
[set_pg_via_master_rule\(2\)](#)

